

Sunyoung Jung

Seoul, South Korea

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Biography

I am a Third-year Ph.D. student at Yonsei University, advised by **Prof. Seong Jae Hwang**, in the **Medical Imaging and Computer Vision (MICV)**. My doctoral research explores the capabilities of **generative models** within the domain of computer vision. I am actively working on problems in **video and image generation, motion transfer in video generation, and diffusion models**.

Education

Ph.D. in Artificial Intelligence Yonsei University Advisor: Seong Jae Hwang	Seoul, South Korea Mar 2025 – Present
M.S. in Electrical and Electronic Engineering Yonsei University Advisor: Dong-Hyun Kim	Seoul, South Korea Mar 2023 – Feb 2025
B.S. in Biotechnology Yonsei University	Seoul, South Korea Mar 2021 – Feb 2023
B.S. in Biomedical Engineering Yonsei University	Wonju, South Korea Mar 2019 – Feb 2023

Research Experience

University of California, Irvine (UCI) <i>Incoming Visiting Scholar</i>	Irvine, CA, USA Sep 2026 – Dec 2026
- Selected for a short-term visiting research appointment focused on generative models and computer vision. - Will conduct collaborative research on controllable video generation with Prof. Alexander C. Berg in the Department of Computer Science.	
Medical Imaging and Computer Vision (MICV), Yonsei University <i>Ph.D. Researcher, Generative Models and Computer Vision</i>	Seoul, South Korea Mar 2025 – Present
- Conduct research on video and image generation, diffusion models, and controllable motion transfer. - Developed attention-head-based motion control for diffusion transformers, resulting in an ECCV 2026 paper as an equal-contribution first author. - Proposed reference-unbiased video motion transfer methods using query-aligned injection and aligned query fusion to reduce reference bias and improve temporal consistency. - Established an industry collaboration with RASCA FX to investigate applications of controllable video generation in real-world VFX production workflows. - Collaborated with Prof. Ming-Hsuan Yang at UC Merced and researchers at Google DeepMind on diffusion-based motion transfer and medical image analysis.	
Medical Imaging Artificial Intelligence Lab, Yonsei University <i>Graduate Researcher and Research Intern, Medical Image Computing</i>	Seoul, South Korea Aug 2022 – Feb 2025
Designed deformation-aware segmentation models robust to MRI motion artifacts, leading to oral presentations at MICCAI 2024 and ISMRM 2024.	

- Applied disentangled representation learning to brain MRI segmentation and through-plane super-resolution across multi-contrast MRI.
- Contributed to research on subcortical segmentation, brain age prediction, motion correction, and MRI super-resolution, with publications at MICCAI, ISMRM, ICMRI, and *Medical Image Analysis*.

Human Brain Function Laboratory, Seoul National University

Research Intern

Seoul, South Korea

Jun 2021 – Sep 2021

- Supported neuroscience research under the supervision of Prof. Chun Kee Chung.

Publications (* Equal Contribution)

Conference Paper

- Controlling Motion Transfer in Diffusion Transformers via Attention Heads
Sunyoung Jung*, Jiwoo Park*, Yoonseok Choi, Kyobin Choo, Ming-Hsuan Yang, and Seong Jae Hwang
ECCV 2026
- QWERTY: Training-Free Motion Control via Query-Warped Video Diffusion Transformers
Kyobin Choo, Youngmin Kim, Hyunkyung Han, Geunrip Park, Chanyoung Kim, **Sunyoung Jung**, and Seong Jae Hwang
ECCV 2026
- Query-Aligned Selective Injection for Mitigating Reference Bias in Motion Transfer
Sunyoung Jung*, Jiwoo Park*, Yeonkyung Lee, Tae Eun Choi, Woojung Han, and Seong Jae Hwang
Under Review
- AQUA: Aligned Query Fusion for Reference-Unbiased and Temporally Consistent Video Motion Transfer
Sunyoung Jung*, Jiwoo Park*, Yeonkyung Lee, Tae Eun Choi, and Seong Jae Hwang
ICML 2026 Workshop - From Frames to Stories (F2S)
- Anatomically Consistent TMJ Disc Segmentation via Semantic Anchoring and Clinical Priors
Dayun Ju, Chanyoung Kim, **Sunyoung Jung**, Hyo-Jung Jung, Chena Lee, Younjung Park, and Seong Jae Hwang
MICCAI 2026
- Interpreting vision transformers via residual replacement model
Jinyeong Kim*, Junhyeok Kim*, Yumin Shim, Joohyeok Kim, **Sunyoung Jung**, and Seong Jae Hwang
NeurIPS 2025
- TESLA: Test-time Reference-free Through-plane Super-resolution for Multi-contrast Brain MRI
Yoonseok Choi, **Sunyoung Jung**, Mohammed A. Al-masni, Ming-Hsuan Yang, and Dong-Hyun Kim
MICCAI 2025, Oral presentation, Top 2.2%
- Attention-guided deep learning model focusing on myelination for predicting brain age using multi contrast MRI
Changmin Ryu, **Sunyoung Jung**, Na-Young Shin, and Dong-Hyun Kim
ISMRM 2025, Oral presentation
- Improving subcortical segmentation in brain MRI using knowledge distillation to enhance robustness against motion artifacts
Changmin Ryu, **Sunyoung Jung**, Yoonseok Choi, and Dong-Hyun Kim
ISMRM 2025
- SUPREM: A Super-Resolution Network for Through-Plane Structure Enhancement Using Disentangled Representation Learning in Multi-Contrast MRI
Yoonseok Choi, **Sunyoung Jung**, Mohammed A. Al-masni, Daniel Kim, and Dong-Hyun Kim
ISMRM 2025

11. Improving subcortical segmentation in brain MRI using knowledge distillation to enhance robustness against motion artifacts
Changmin Ryu, **Sunyoung Jung**, Yoonseok Choi, and Dong-Hyun Kim
ICMRI 2025, Best Trainee Scientific Awards Poster Presentation (Gold)
 12. Deformation-Aware Segmentation Network Robust to Motion Artifacts for Brain Tissue Segmentation using Disentanglement Learning
Sunyoung Jung, Yoonseok Choi, Mohammed A. Al-masni, Minyoung Jung, and Dong-Hyun Kim
MICCAI 2024
 13. Brain Tissue Segmentation Robust to motion artifacts using Deformation-Aware Network
Sunyoung Jung, Yoonseok Choi, Mohammed A. Al-masni, and Dong-Hyun Kim
ISMRM 2024, Oral presentation
 14. Investigation of electrical conductivity changes during functional activity of the brain via phase-based MR-EPT: Preliminary observation
Kyu-Jin Jung, Chuanjiang Cui, Jae-Hun Lee, Jun-Hyeong Kim, Kyoung-Jin Park, SooHyoung Lee, **Sunyoung Jung**, Dong-Hyun Kim
ISMRM 2023, Oral presentation
 15. 3D Lesion Generation Model considering Anatomic Localization to Improve Object Detection in Limited Lacune Data
Daniel Kim, Jae-Hun Lee, Mohammed A. Al-masni, Jun-ho Kim, Yoonseok Choi, Eun-Gyu Ha, **Sunyoung Jung**, Dong-Hyun Kim
ISMRM 2023
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Journal Paper

1. Motion Artifact-Informed Brain Tissue Segmentation Network via Disentanglement Learning
Sunyoung Jung, Yoonseok Choi, Mohammed Al-masni, Min-Young Jung, Dong-Hyun Kim
Medical Image Analysis (MedIA), Under Review, 2025, impact factor 11.8
2. Test-time Reference-free Through-plane Super-resolution Network for Multi-contrast Brain MRI via Disentangled Representations
Yoonseok Choi, **Sunyoung Jung**, Gayoon Choi, Mohammed A. Al-masni, Kyu-Jin Jung, Wei-Ting Chen, Ming-Hsuan Yang, and Dong-Hyun Kim
Medical Image Analysis (MedIA), Under Review, 2025, impact factor 11.8
3. Unsupervised learning for motion correction and assessment in brain magnetic resonance imaging using severity-based regularized cycle consistency
Seuk Kim*, Mohammed A. Al-masni*, Seul Lee, **Sunyoung Jung**, Kyu-Jin Jung, Chuanjiang Cui, Sung-Min Gho, Young Hun Choi, and Dong-Hyun Kim
Engineering Applications of Artificial Intelligence, 2025, impact factor 8.0

Awards and Honors

1. Merit based scholarship Science and Engineering Doctor, Korea, 2026
2. Seoul Future Foundation Scholarship, Korea, 2026
3. ICMRI 2025 Best Trainee Scientific Awards Poster Presentation, Gold, Korea, 2025
4. ISMRM 2024 stipend, Korean Society of Magnetic Resonance in Medicine, Korea, 2024
5. Academic Achievement Award, Yonsei University, Korea, 2019

Professional Experience

Conference Reviewer

<i>The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)</i>	2026
<i>The European Conference on Computer Vision (ECCV)</i>	2026
<i>Neural Information Processing Systems (NeurIPS)</i>	2026

Teaching Assistant

<i>Introduction to Deep Learning and Application, Yonsei University</i>	<i>spring 2026</i>
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